

Assignment 6: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A06_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
knitr::opts_chunk$set(tidy = TRUE, tidy.opts = list(width.cutoff = 20))
library(rvest)
library(tidyverse)
getwd()
```

```
## [1] "C:/Users/24169/Desktop/e872/872_spring26"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Set the year to 2024
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
# 2
website <- read_html("https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024")
glimpse(website)
```

```
## List of 2
## $ node:<externalptr>
## $ doc :<externalptr>
## - attr(*, "class")= chr [1:2] "xml_document" "xml_node"
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- Contact Person
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
# 3
introduce_first <- website %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()
introduce_first

## [1] "Durham"

introduce_second <- website %>%
  html_nodes("div+ table tr:nth-child(4) td:nth-child(2)") %>%
  html_text()
introduce_second

## [1] "Reginald Hicks"

introduce_third <- website %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()
introduce_third

## [1] "Municipality"

introduce_fourth <- website %>%
  html_nodes("th~ td+ td") %>%
  html_text()
introduce_fourth

## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It’s likely you won’t be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: “Jan”, “May”, “Sept”, “Feb”, etc. . .

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

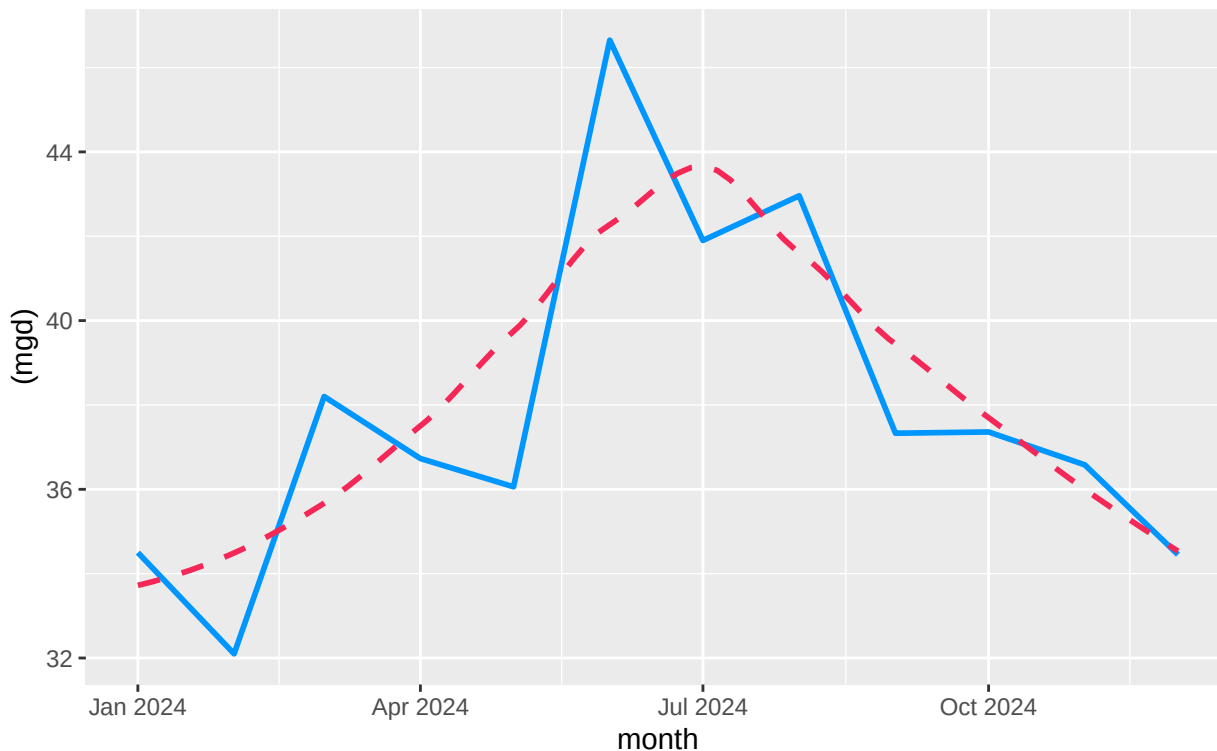
```
# 4
report_first <- data.frame(Water_System_Name = rep(introduce_first,
  times = 12), Contact_Person = rep(introduce_second,
  times = 12), Ownership = rep(introduce_third,
  times = 12), Maximum_Day_Use = as.numeric(introduce_fourth),
  Year = rep(2024,
    times = 12),
  Month = c(1, 5, 9,
    2, 6, 10, 3,
    7, 11, 4, 8,
    12))
report_first <- report_first %>%
  mutate(Date = ym(paste(Year,
    "-", Month)))
str(report_first)

## 'data.frame':    12 obs. of  7 variables:
## $ Water_System_Name: chr  "Durham" "Durham" "Durham" "Durham" ...
## $ Contact_Person   : chr  "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" ...
## $ Ownership        : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Maximum_Day_Use  : num  34.5 36.1 37.3 32.1 46.6 ...
## $ Year              : num  2024 2024 2024 2024 2024 ...
## $ Month             : num   1  5  9  2  6 10  3  7 11  4 ...
## $ Date              : Date, format: "2024-01-01" "2024-05-01" ...

# 5
p1 <- ggplot(data = report_first,
  aes(x = Date, y = Maximum_Day_Use)) +
  geom_line(linewidth = 1,
    color = "#0096FF") +
  geom_smooth(method = "loess",
    se = FALSE, linetype = "dashed",
    linewidth = 1,
    color = "#F52757") +
  theme_gray() + labs(title = "Maximum Day Use Of Water",
    subtitle = "Data In 2024, From Durham",
    x = "month", y = "(mgd)")
p1
```

Maximum Day Use Of Water

Data In 2024, From Durham



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:
- Creates a URL pointing to the LWSP for that PWSID for the given year
 - Creates a website object and scrapes the data from that object (just as you did above)
 - Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
 - Returns the dataframe as the function’s output

```
# 6.
scrape <- function(year,
  facility) {
  target <- read_html(paste0("https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=",
    facility, "&year=",
    year))
  tag_first <- "div+ table tr:nth-child(1) td:nth-child(2)"
  tag_second <- "div+ table tr:nth-child(4) td:nth-child(2)"
  tag_third <- "div+ table tr:nth-child(2) td:nth-child(4)"
  tag_fourth <- "th~ td+ td"
  universal_first <- target %>%
    html_nodes(tag_first) %>%
    html_text()
  universal_second <- target %>%
    html_nodes(tag_second) %>%
    html_text()
  universal_third <- target %>%
    html_nodes(tag_third) %>%
    html_text()
  universal_fourth <- target %>%
    html_nodes(tag_fourth) %>%
    html_text()
  report_all <- data.frame(Water_System_Name = rep(universal_first,
```

```

    times = 12),
  Contact_Person = rep(universal_second,
    times = 12),
  Ownership = rep(universal_third,
    times = 12),
  Maximum_Day_Use = as.numeric(universal_fourth),
  Year = rep(year,
    times = 12),
  Month = c(1,
    5, 9, 2,
    6, 10, 3,
    7, 11, 4,
    8, 12)) %>%
  mutate(Date = ym(paste(Year,
    "-", Month)))
return(report_all)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

# 7
report_second <- scrape(2020,
  "03-32-010")
p2 <- ggplot(data = report_second,
  aes(x = Date, y = Maximum_Day_Use)) +
  geom_line(linewidth = 1,
    color = "#0096FF") +
  geom_smooth(method = "loess",
    se = FALSE, linetype = "dashed",
    linewidth = 1,
    color = "#F52757") +
  theme_gray() + labs(title = "Maximum Day Use Of Water",
    subtitle = "Data In 2020, From Durham",
    x = "month", y = "(mgd)")
p2

```

Maximum Day Use Of Water

Data In 2020, From Durham



```
str(report_second)
```

```
## 'data.frame':    12 obs. of  7 variables:
## $ Water_System_Name: chr  "Durham" "Durham" "Durham" "Durham" ...
## $ Contact_Person   : chr  "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" ...
## $ Ownership        : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Maximum_Day_Use  : num  36 37 41.7 32 40.6 ...
## $ Year             : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ Month            : num  1 5 9 2 6 10 3 7 11 4 ...
## $ Date             : Date, format: "2020-01-01" "2020-05-01" ...
```

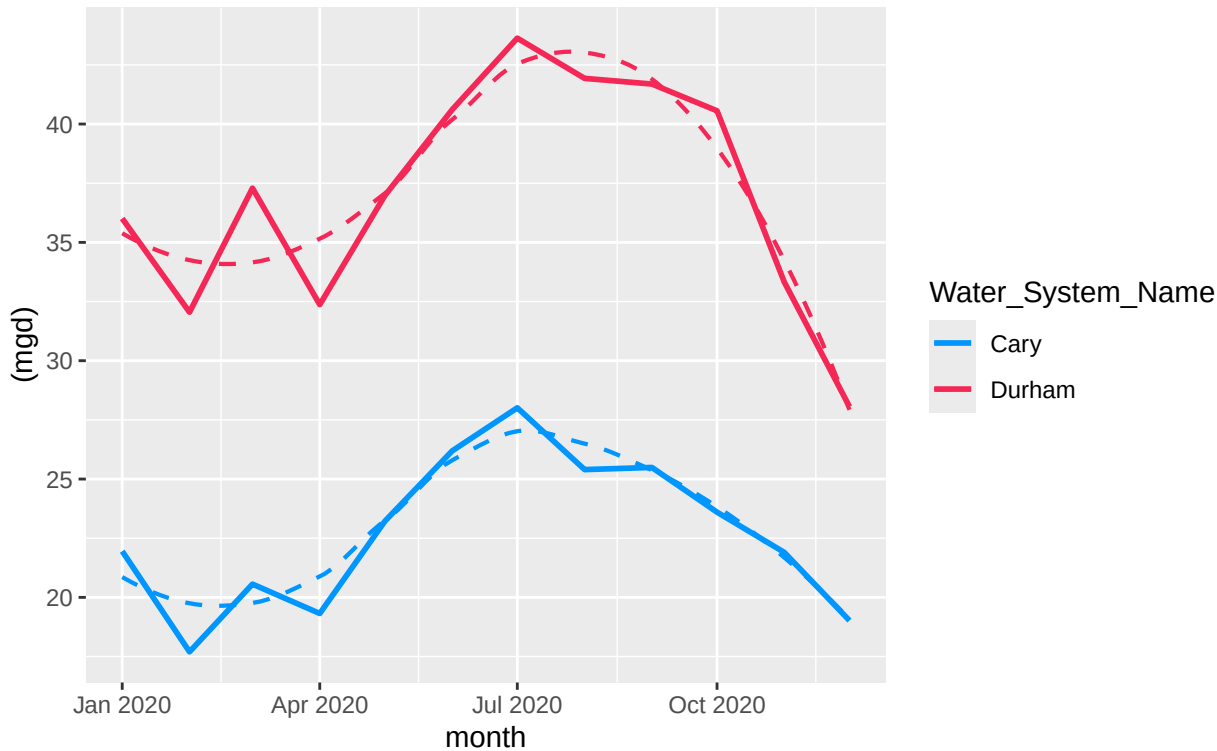
- Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
# 8
report_third <- scrape(2020,
  "03-92-020")
comparison_first <- bind_rows(report_second,
  report_third)
p3 <- ggplot(data = comparison_first,
  aes(x = Date, y = Maximum_Day_Use,
    color = Water_System_Name)) +
  geom_line(linewidth = 1) +
  geom_smooth(method = "loess",
    se = FALSE, linetype = "dashed",
    linewidth = 0.75) +
  scale_color_manual(values = c("#0096FF",
    "#F52757")) +
  theme_gray() + labs(title = "Maximum Day Use Of Water",
    subtitle = "Data In 2020, From Cary And Durham",
    x = "month", y = "(mgd)")
```

p3

Maximum Day Use Of Water

Data In 2020, From Cary And Durham

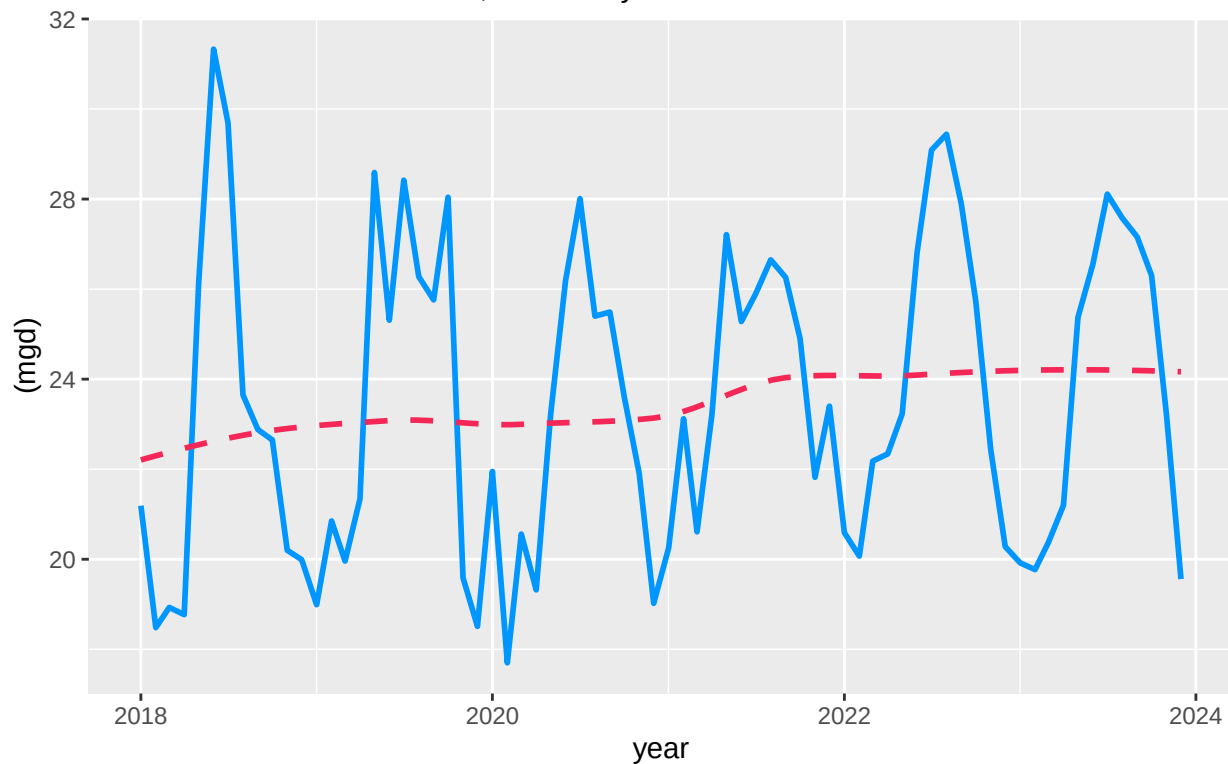


- Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
# 9
history <- c(2018, 2019,
             2020, 2021, 2022,
             2023)
location <- "03-92-020"
comparison_second <- map2(history,
                           location, scrape) %>%
  bind_rows()
p4 <- ggplot(data = comparison_second,
             aes(x = Date, y = Maximum_Day_Use)) +
  geom_line(linewidth = 1,
            color = "#0096FF") +
  geom_smooth(method = "loess",
              se = FALSE, linetype = "dashed",
              linewidth = 1,
              color = "#F52757") +
  theme_gray() + labs(title = "Maximum Day Use Of Water",
                      subtitle = "Data Between 2018 And 2023, From Cary",
                      x = "year", y = "(mgd)")
p4
```

Maximum Day Use Of Water
Data Between 2018 And 2023, From Cary



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time?

Answer: In general, the water use during summer is more than that during winter. Cary might witness increasing water use over time. However, this trend is not significant.